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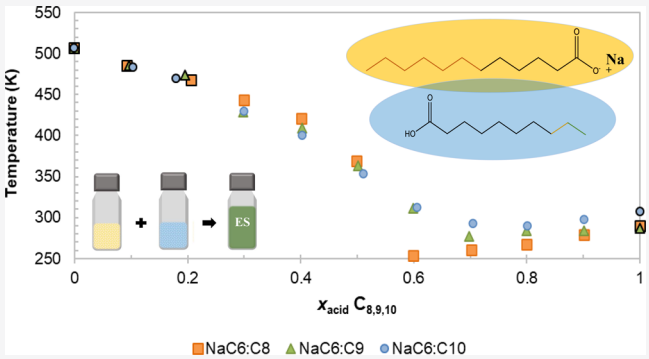
# Sodium Hexanoate and Dodecanoate Salt-Based Eutectic Solvents: Density, Viscosity, and Kamlet–Taft Parameters

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**ABSTRACT:** Eutectic solvents (ESs) are a relatively new class of solvents that have been arousing a lot of attention due to their easy preparation and great tunability of their thermophysical properties, which has been granting their application in a wide range of areas. The aim of this work is to present a new ES based on the mixture of an organic sodium salt, sodium hexanoate and dodecanoate, and a long alkyl chain carboxylic acid, octanoic, nonanoic, and decanoic acids, and to infer about the effect of introducing a sodium atom. The solid–liquid phase diagrams of these new sodium salt-based ES were determined to evaluate the liquid window where they can be confidently used as solvents. Other important properties of solvents have also been studied, and in particular, density and viscosity, at atmospheric pressure and in the temperature range between 293.15 and 353.15 K, were measured. In addition, the polarity of these new sodium salt-based deep eutectic solvents (DESs) was evaluated in terms of the Kamlet–Taft $\alpha$, $\beta$, and $\pi^*$ and $E_T$N parameters. The obtained results were compared with those of other commonly used hydrophilic and hydrophobic DESs and conclusions were taken.




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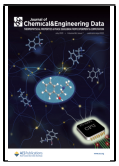
## 1. INTRODUCTION

Solvents are omnipresent in chemical processes and are also one of the major contributors to industrial waste streams since most of the conventional organic solvents commonly used in industry are volatile, toxic, corrosive, and flammable.[1] The development of new eco-friendly solvents, meeting important green chemistry requirements, such as atom economy, less hazardous chemical synthesis, use of materials from renewable resources, and design for degradation, represents an important challenge for the development of sustainable technologies.[2] Deep eutectic solvents (DESs) are eutectic mixtures of Lewis acids and bases that present a large decrease in the melting point of the mixture compared to that of the ideal mixture.[3,4] This large deviation from the ideal mixture behavior is attributed to the complex network of hydrogen bonds between the two, or more, components of the DESs and allows us to considerably extend the area of ($T$, $x$) conditions where a liquid mixture is attained at room temperature, thus affording their use as solvents. Other eutectic solvents (ESs) with smaller deviations from the ideal mixtures have also been successfully used in several applications.

Since the first publication in 2004,[4] DESs have been widely used in a variety of fields such as catalysis, organic synthesis, electrochemical devices, solar technology, and so forth.[5] More recently, DESs and ESs have also found other applications such as in extraction, separation, and purification processes, for example, in selective isolation and recovery of metals,[6] purification of compounds from natural products,[7] purification of fuels,[8] azeotrope breaking,[9] and purification of water,[10,11] to mention a few.

A wide variety of compounds have been used to prepare eutectic solvents, either DES or ES. In particular, natural compounds such as an amino acid,[12] a carbohydrate,[13] an alcohol,[14] and a carboxylic acid,[15] among others have been extensively used as a hydrogen-bond donor (HBD) and organic salts, such as ammonium and phosphonium salts, with particular emphasis on choline chloride, have been used as a hydrogen-bond acceptor (HBA). Depending on the judicious selection of the compounds, in terms of molecular structure and molar ratio, important properties for the envisaged application can be attained. For example, depending on the chosen individual compounds, DESs can display negligible environmental and economic impact due to their non-flammability, biodegradability, ease of preparation with no further purification, low volatility, and low prices.[16,17] In this perspective, although most DESs and ESs reported are hydrophilic,[18,19] hydrophobic DESs and ESs have lately been proposed and largely explored.

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**Table 1. CAS Registry Number, Mass Fraction Purity, and Chemical Source**

| chemical name | Cas reg. no | source | mass fraction purity | purification method | analysis method |
|---|---|---|---|---|---|
| sodium hexanoate | 10051-44-2 | Acros Organics | 0.99 | none | |
| sodium dodecanoate | 629-25-4 | TCI | 0.97 | none | |
| octanoic acid | 124-07-2 | Acros Organics | 0.99 | none | |
| nonanoic acid | 112-05-0 | Sigma-Aldrich | 0.97 | none | |
| decanoic acid | 334-48-5 | Alfa Aesar | 0.99 | none | |
| 2,6-dichloro-4-(2,4,6-triphenyl-N-pyridino)-phenolate | 10081-39-7 | Fluka | >0.97 | none | |
| 4-nitroaniline | 100-01-6 | Sigma-Aldrich | 0.99 | none | |
| N,N-diethyl-4-nitroaniline | 2216-15-1 | Frinton Laboratories | | none | |

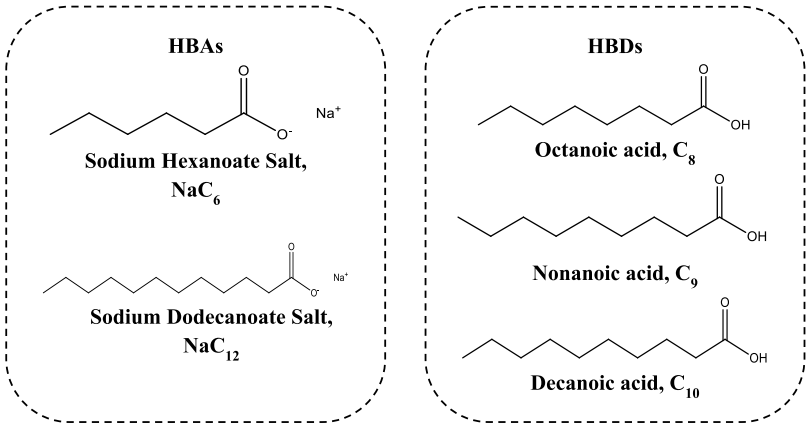



**Figure 1.** Chemical structures and acronyms of the HBAs and HBDs used to prepare the sodium salt-based DESs studied in this work.

In our group, the development of hydrophobic ESs has been carried out combining two, or more, hydrophobic compounds, such as terpenes and fatty acids, to be essentially used in the development of water purification technologies.[10,20−23] Other groups have been also contributing with DESs based on the combination of fatty acids and ammonium- and phosphonium-bearing long alkyl chains.[20] Recently, we proposed a new ES composed of a long chain fatty acid and sodium salts derived from long alkyl chain fatty acids. Abbott *et al.*[24−26] combined alkali metal salts, namely, based on sodium, potassium, magnesium, and calcium salts, with organic compounds but only a limited number formed homogeneous liquids at room temperature, all of them with glycerol. These authors concluded that none of these mixtures showed eutectic behavior, but the obtained liquids showed physical properties similar to those of DESs. Sedghamiz and Raeissi[26] have also reported DESs based on sodium salts (sodium chloride, sodium bromide, and sodium iodide) and ethylene glycol and thermophysical properties, such as density, viscosity, and refractive index, as well as the effect of different amounts of water on these properties.

The main goal of the present work is to characterize ESs based on sodium alkanoate salts with long alkyl chains combined with several fatty acids envisaging their application in separation processes. For that purpose, mixtures of one HBA, hexanoate sodium salt ($NaC_6$) and dodecanoate sodium salt ($NaC_{12}$), and one natural fatty acid, such as HBD, octanoic acid ($C_8$), nonanoic acid ($C_9$), and decanoic acid ($C_{10}$), were screened for the formation of metal-based DESs. Solid−liquid phase diagrams at ambient pressure were measured to determine the temperature-composition window at which these solvents are homogeneous liquids at room temperature. Density and viscosity data of these new ESs were measured. Polarity, in terms of the normalized polarity ($E_TN$) and the Kamlet−Taft parameters, $\alpha$, $\beta$, and $\pi^*$, was also evaluated. The measured properties were compared with those of other ESs and discussed.

## 2. EXPERIMENTAL SECTION

**2.1. Materials.** HBAs, sodium hexanoate ($C_6H_{11}NaO_2$) (CAS 10051-44-2) with 99% purity and sodium dodecanoate ($C_{12}H_{23}NaO_2$) (CAS 629-25-4) with 97% purity, were purchased from Acros Organics and TCI, respectively. For the three HBDs, octanoic acid ($C_8H_{16}O_2$) (CAS 124-07-2) with 99% purity, nonanoic acid ($C_9H_{18}O_2$) (CAS 112-05-0) with 97% purity, and decanoic acid ($C_{10}H_{20}O_2$) (CAS 334-48-5) with 99% purity were purchased from Acros Organics, Sigma, and Alfa Aesar, respectively. 2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridinio)-phenolate (Reichardt's betaine dye 33), 4-nitroaniline, and *N*,*N*-diethyl-4-nitroaniline were purchased from Fluka (≥97% mass fraction purity), Sigma Aldrich (≥99% mass fraction purity), and Frinton Laboratories, respectively. All chemical compounds are listed in Table 1 and were used as received. Chemical structures of all HBAs and HBDs used for synthesis of the sodium salt-based DESs studied in this work are presented in Figure 1, while the chemical structures of the solvatochromic probes are presented in Figure S19 of the Supporting Information. Ultrapure, double distilled water passed through a reverse osmosis system and further treated with a Milli-Q plus water purification apparatus was used.

**2.2. Apparatuses and Procedures.** *2.2.1. Preparation Methodology of ESs.* The sodium hexanoate and dodecanoate salts were dried under vacuum and mild heating overnight before they were used. The water content of these ESs is reported in Table 2 as well as the water content of the octanoic, nonanoic, and decanoic acids, which were used as received.


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**Table 2. Summary Properties for the Pure Compounds and Their ES Mixtures at a Chosen Composition Close to the Eutectic Point**[a]

| compounds | $x_{acid}$ | MW (g/mol) | water content (wt %) | melting temperature (K) | literature melting temperature (K) |
|---|---|---|---|---|---|
| NaC6 | | | 0.035 | 493.00 | |
| NaC12 | | | 0.022 | 518.15 | 517.15[28] |
| C8 | | | 0.004 | 289.65 | 289.65[33] −289.85[30] |
| C9 | | | 0.006 | 285.65 | 285.55[33] −285.65[31] |
| C10 | | | 0.019 | 304.75 | 304.65[34] −304.75[30] |
| $NaC_6/C_8$ | 0.7 | 142.19 | 0.082 | 253.65 | |
| $NaC_6/C_9$ | 0.7 | 151.53 | 0.095 | 273.55 | |
| $NaC_6/C_{10}$ | 0.7 | 160.89 | 0.038 | 290.15 | |
| $NaC_{12}/C_8$ | 0.8 | 159.83 | 0.101 | 268.85 | |
| $NaC_{12}/C_9$ | 0.8 | 171.04 | 0.041 | 265.05 | |
| $NaC_{12}/C_{10}$28 | 0.8 | 182.27 | 0.028 | 296.15 | |

[a]The compositions $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 were chosen, for comparison purposes, for NaC6- and NaC12-containing ES families, respectively. The water content of the ES mixtures used to measure densities and viscosities is also reported.

Mixtures of sodium salts and carboxylic acids with long alkyl chains were prepared by the addition of the acid to the dried sodium salt in a sealed glass vial. Each one of the ES components was carefully weighed, using an analytical high-precision balance Mettler Toledo scale (New Classic MS205DU) with an uncertainty of $\pm 10^{-4}$ g and then slowly heated up with constant stirring, using a heating-stirring plate (Heidolph, MR Hei-Tec), until a clear liquid phase was observed. The heat was turned off and the mixtures were allowed to slowly cool down to room temperature. The water content of the dried components and the prepared ES was determined by Karl Fischer titration (Metrohm 870 KF Titrino Plus).

*2.2.2. FTIR Spectra Determination.* To rule out the possibility of the formation of anhydrides and/or ionic exchange, Fourier transform infrared (FTIR) spectroscopy was used. As an example, the FTIR spectrum of NaC6/C8 at $x_{acid}$ = 0.6 was acquired using a PerkinElmer Spotlight 200i FTIR microscope system as is shown in Figure S18 of the Supporting Information. The spectrum was collected using 8 scans, from 450 to 4000 $cm^{-1}$, with a resolution of 4 $cm^{-1}$. In addition, FTIR was also used to confirm the formation of the ES through the establishment of hydrogen bonds.

*2.2.3. Solid−Liquid Phase Diagram Determination.* Two methods were used to measure the solid−liquid phase diagrams at atmospheric pressure: turbidimetry, by the visual observation of turbidity, was used for temperatures below 443.15 K and differential scanning calorimetry (DSC) was used for temperatures above 443.15 K, where it was not possible to use the turbidity measurements. For the turbidimetric measurements, approximately, 5 g of each ES mixtures with different compositions were prepared, according to the preparation methodology described above, in a glass flask of 10 mL of total volume. After obtaining the fully homogeneous liquid mixture, the heating was turned off, and when the first solid was detected, the temperature was registered. The temperature was measured with a calibrated PT100 probe with a precision of ±0.1 K connected to a multimeter Keithley 199 System DMM/Scanner that was calibrated with high-accuracy mercury thermometers (0.01 K). For mixtures that did not reach the homogeneous liquid state at 443.15 K, the mixture is sealed inside a pan and placed inside DSC. The DSC thermograms are provided in Figures S1−S17 of the Supporting Information.

For the DSC experiments, a Q200 (TA Instruments) DSC was used. Samples were weighed (∼9−12 mg) with a precision of $\pm 10^{-4}$ g and hermetically sealed in aluminum pans. For all samples, DSC experiments were carried out at a heating rate of 10 °K $min^{-1}$ in a large temperature window, from 203.15 to 543 K, to obtain broader insights into the ES thermal profiles.

*2.2.4. Thermophysical Properties.* Measurements of viscosity and density properties of these sodium salt-based DESs were carried out in the temperature range from 293.15 up to 353.15 K and at atmospheric pressure, using an Anton Paar (model SVM 3000) automated rotational Stabinger viscometer-densimeter, with a temperature uncertainty of ±0.01 K. The relative uncertainty of the dynamic viscosity is ±0.25%, and the absolute uncertainty of the density is ±0.0005 $g \cdot cm^{-3}$.

*2.2.5. Solvatochromic Probes.* A predetermined amount of each solvatochromic probes was weighed using a Mettler Toledo scale (New Classic MS205DU) with an accuracy of $\pm 10^{-4}$ g and dissolved in dichloromethane in precleaned glass

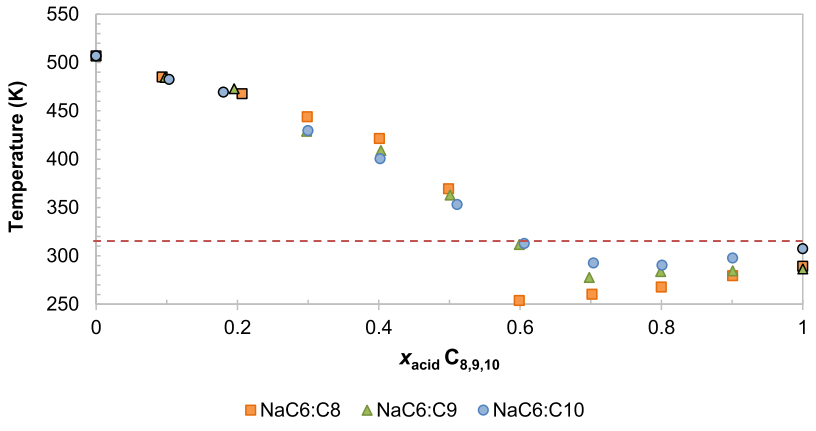



**Figure 2.** Experimental ($T$, $x$) solid−liquid phase diagram for the sodium hexanoate salt-based DES with each acid prepared in this work. DSC data are outlined with a black line, while the remaining data were obtained by turbidimetry.



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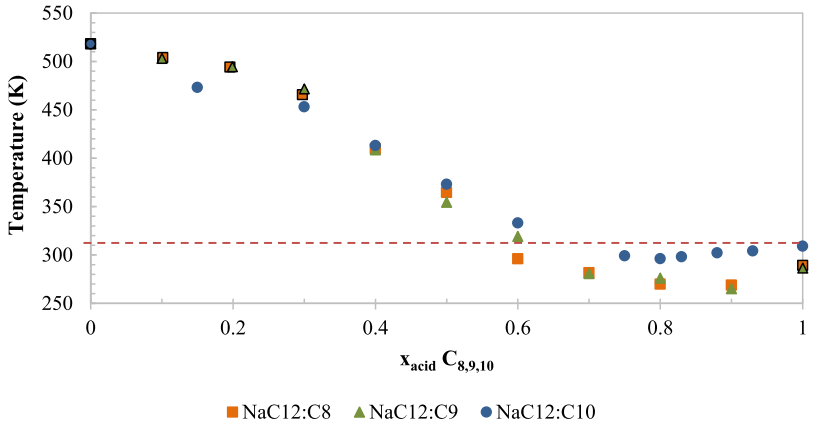



**Figure 3.** Experimental ($T$, $x$) solid−liquid phase diagram for the sodium dodecanoate salt-based DES prepared in this work. DSC data are outlined with a black line, while the remaining data were obtained by turbidimetry. The data regarding $NaC_{12}/C_{10}$ were taken from Florindo *et al.*[27]

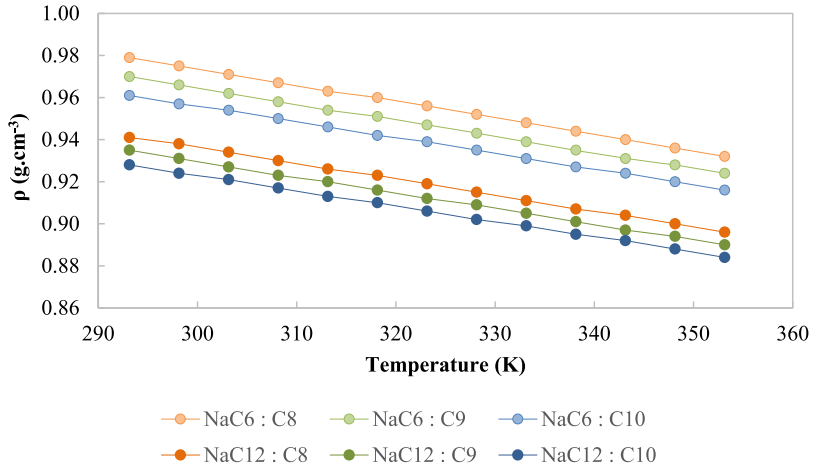



**Figure 4.** Experimental densities of the two families of dried sodium salt-based DESs, at $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 for NaC6- and NaC12-based ESs, as a function of temperature and at atmospheric pressure.

vials. The dichloromethane of probe solution was evaporated using a gentle stream of high-purity nitrogen gas. An amount of ES was directly added to the probe solution, thoroughly mixed, and then transferred to 1 cm light path quartz cuvette. A Shimadzu UV-1800 spectrophotometer (UV/vis) was used for acquisition of the UV−vis molecular absorbance data. All spectroscopic measurements were performed in triplicate. All UV−vis spectra are provided in Figures S20−S37 of the Supporting Information.

## 3. RESULTS AND DISCUSSION

**3.1. Solid−Liquid Phase Diagram.** The melting points of each one of the prepared ES mixtures at atmospheric pressure were determined according to the experimental procedure described above. The ($T$, $x$) solid−liquid phase diagrams for the ES based on the sodium hexanoate and sodium dodecanoate salts are shown in Figures 2 and 3, respectively, and the data are reported in Table S1 in the Supporting Information. The data previously reported by us[27] for $NaC_{12}/C_{10}$ were included for completeness and comparison.

The melting points of the pure compounds, measured in this work using DSC, can be compared with those reported in ref[28−30] at Table 2. It can be concluded that there is a good agreement between the measured data and the literature data and the small deviations are probably due to impurities or experimental uncertainty. For the NaC6 salt, no melting temperature was found in the open literature.

Both phase diagrams shown in Figures 2 and 3 have a similar behavior, being the most prominent difference in the range of compositions at which the eutectic point occurs: in the case of NaC6/acids, it occurs at $x_{acid}$ = 0.6−0.7, while in the case of NaC12/acids, it occurs at around $x_{acid} \approx 0.80$. This shows that the HBA has a greater impact in the determination of the eutectic point than the HBD. This same trend was already observed in the literature. For example, Coutinho *et al.*[21] prepared numerous DESs containing choline chloride as an HBA and alcohols and carboxylic acids and showed that the eutectic point composition does not change significantly with the HBD. The same trend was also observed by Hou *et al.*[32] for the DES containing choline chloride with diverse components such as urea, glycerol, imidazole, or phenol.

The dashed red line in Figures 2 and 3 at 298.15 K represents the room temperature and it was drawn to highlight the range of compositions at which a liquid phase is attained through the formation of a eutectic system. Generally, it can be observed that for both NaC6 and NaC12 DES families, the largest decrease in the melting point is observed for the ES containing *n*-octanoic acid. NaC6/C8 DES presents the greatest gain in terms of the working temperature range, since there is a large temperature difference, a drop of 36 K, between the temperature of the eutectic point (253.65 K) and


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the lowest melting temperature of the pure components. In the case of the NaC12-based DES family, the gain in terms of liquid window is not as expressive, and both NaC12/C8 and NaC12/C9 show a similar decrease in the melting temperature (around 298.15 K) compared to the melting temperature of the corresponding acids. These temperature differences between the melting temperature of the pure compounds and their NaC6- and NaC12-based ESs can be observed in Table 2. Since in this work, the eutectic points of the prepared ES mixtures could not be determined, the compositions $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 were chosen, for comparison purposes, for NaC6- and NaC12-containing ES families.

**3.2. Thermophysical Properties: Density and Viscosity.** Density and viscosity are very important properties of solvents since they can influence the mass transport phenomena, affecting their application. Density and viscosity of the dried sodium salt-based ES, at $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 for NaC6- and NaC12-based ESs, were measured within the range of 293.15 to 353.15 K at atmospheric pressure. The experimental density results for the dried ES as a function of temperature are presented in Figure 4 and listed in Table S2 of the Supporting Information. As usual, a linear temperature dependence of density was used

$$\rho = a + bT \quad (1)$$

where $\rho$ corresponds to density in g·cm$^{-3}$, $T$ is the temperature in K, and $a$ and $b$ are the fitting parameters, which are presented in Table 3.

**Table 3. Fitted Values of Parameters, $a$ and $b$, in eq 1, and Correlation Coefficient Values, $R^2$, for the Two Families of Sodium Salt-Based ESs**

| sodium salt-based DESs | | a (g·cm$^{-3}$) | b × 10$^4$ (g·cm$^{-3}$·K) | $R^2$ |
|---|---|---|---|---|
| $NaC_6$ | $C_8$ acid | 1.2066 | −7.77 | 1 |
| | $C_9$ acid | 1.1941 | −7.65 | 1 |
| | $C_{10}$ acid | 1.1815 | −7.51 | 1 |
| $NaC_{12}$ | $C_8$ acid | 1.1628 | −7.55 | 1 |
| | $C_9$ acid | 1.1525 | −7.44 | 1 |
| | $C_{10}$ acid | 1.1424 | −7.31 | 1 |

As expected, for all dried sodium salt-based ESs, the density decreases linearly with temperature in the whole studied temperature range. Two distinct groups of densities, with similar trends, corresponding to the two families of ESs containing sodium salts of different alkyl chains, can be observed in Figure 4. The family of NaC6-based ESs presents higher densities when compared to the family of NaC12-based ESs, ranging at 298.15 K from 0.957 to 0.975 and 0.924 to 0.938 g·cm$^{-3}$, respectively. Keep in mind that these two families have different compositions, as for NaC6/acids, the composition of $x_{acid}$ = 0.7 was taken, while for NaC12/acids, the composition of $x_{acid}$ = 0.8 was considered. This can probably be the explanation for the large difference in densities for these two families of ESs. Comparing these results with the results for other DESs in the literature at 298.15 K, it can be concluded that generally the densities of these sodium salt-based DESs are lower than the most hydrophilic DESs (around 1.150 g·cm$^{-3}$)[34,35] but very similar to the reported hydrophobic ammonium salt-based DES (0.889−0.942 g·cm$^{-3}$).[20]

Fixing the HBA, in the range of temperatures studied, the density follows the same order for the two ES families: octanoic acid > nonanoic acid > decanoic acid, where the latter presents the lowest densities. The decrease in the density values with the increase in the alkyl chain of both HBA and HBD verified in this work has also been reported by other authors. Van Osch *et al.*[20] studied the densities for N4444Cl/C10 (1:2) and N8888Cl/C10 (1:2) DESs and showed that at 298.15 K, density values range from 0.917 to 0.889 g·cm$^{-3}$, respectively. Our group reported DESs based on the mixture of two fatty acids and also verified that the density values decreased with the increase in alkyl chains of the fatty acids.[11]

In what concerns the HBD, the effect of the size of the HBA in ESs containing C10 can also be inferred, using the results obtained in this work for NaC6/C10 at $x_{acid}$ = 0.7 with those in the literature at the composition of 1 HBA/2 HBD. At 298.15 K, an increasing density order can be established in agreement with the decrease in the size of HBA/N8888Cl/C10 (0.889 g·cm$^{-3}$) < C12/C10 (0.894 g·cm$^{-3}$) < N4444Cl/C10 (0.917 g·cm$^{-3}$) < NaC6/C10 (0.957 g·cm$^{-3}$).

The thermal expansion coefficients can be determined from density data using the following equation

$$\alpha_p = -\frac{1}{\rho}\left(\frac{\delta\rho}{\delta T}\right)_p \quad (2)$$

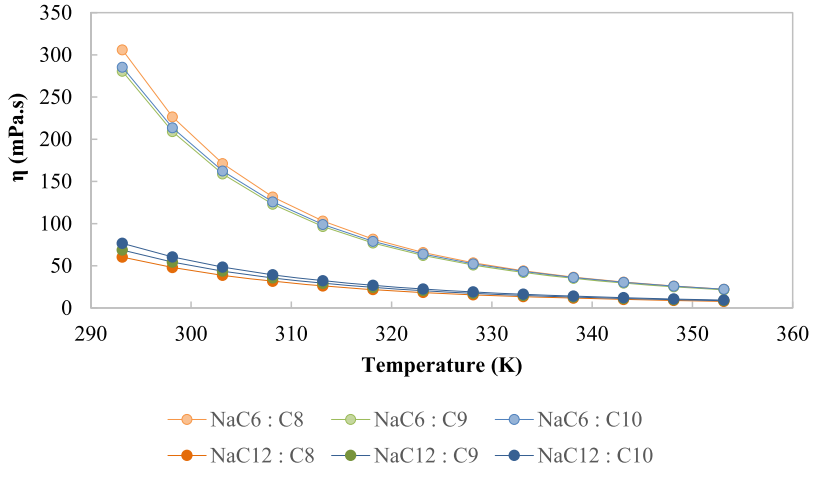



**Figure 5.** Experimental viscosities of the two families of dried sodium salt-based ESs studied in this work, at $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 for NaC6- and NaC12-based ESs, as a function of temperature and at atmospheric pressure.



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where $\rho$ is the density in g·cm$^{-3}$, $T$ is the temperature in K, and $p$ is the pressure in MPa.

The values of the thermal expansion coefficients for the two sodium salt-based DES families studied in this work are presented in Table S3 in the Supporting Information. It can be observed that for each temperature, the thermal expansion coefficients have exactly the same values for both families of DESs with the same HBD and that within each family, the values decrease with the increase in the alkyl chain length of the HBD.

The experimental viscosity results for the dried ES as a function of temperature are presented in Figure 5 and listed in Table S4 in the Supporting Information. The experimental viscosity values were fitted as a function of temperature, using the Vogel−Fulcher−Tammann (VFT) model

$$\ln \eta = A_\eta + \frac{B_\eta}{(T - C_\eta)} \tag{3}$$

where $\eta$ is the viscosity in mPa·s, $T$ is the temperature in K, and $A_\eta$, $B_\eta$, and $C_\eta$ are adjustable parameters. The adjustable parameters of the VFT model were determined from the fitting of the experimental viscosity values with temperature and are listed in Table S5 of the Supporting Information.

As expected, with increasing temperature, an exponential decrease in viscosity was observed for all sodium salt-based ESs. It can also be observed that the viscosity values within each family of sodium salt-based ESs are very similar. However, a large difference in viscosity exists between the two families of ESs, with NaC12 salt-based DESs presenting lower viscosities than those containing the NaC6 salt. At 298.15 K, viscosities vary from 209 to 226.5 mPa·s for the family of NaC6-based ESs, while for the NaC12-based ESs, they vary between 47.97 and 60.48 mPa·s. Although it is important to keep in mind that viscosities were measured at different compositions for both families, this behavior clearly indicates that viscosity is controlled by the HBA.

Generally, comparing the viscosity values obtained in this work for both families of sodium salts-based ESs (between 209.0 and 213.6 mPa·s for NaC6-based ESs and between 47.97 and 60.48 mPa·s for NaC12-based ESs) with those of other DESs reported in the literature, at 298.15 K, it can be observed that the viscosity of both families is well below that reported for ChCl/urea (1:2) (1571 mPa·s).[34] However, the viscosity values of NaC6-based ESs are similar to that of hydrophobic ammonium salt-based DESs (265.26 mPa·s).[20] On the other hand, the viscosity of NaC12-based ESs is of the same order of magnitude of that obtained for ESs composed by menthol/C10 (20.32 mPa·s).[36]

The density and viscosity of other sodium salt-based DESs, such as NaCl/ethylene glycol (1:16), NaBr/ethyleneglycol (1:6), and NaI/ethylene glycol (1:4), have also been reported and are listed in Table S6 of the Supporting Information.[26] Despite the much higher proportions of HBAs present in these DESs than in the sodium alkanoate-based ESs studied in this work, a comparison can be established. At 293 K, although the densities for all these three DESs are higher than the unit and thus higher than those measured for both families of the sodium alkanoate ESs, the viscosities present values between 32.36 and 81.23 mPa·s,[26] in the same order of magnitude of NaC12-based ESs studied in this work.

Another interesting comparison can be made using the data for DESs exclusively composed of long chain alkyl fatty acids,[11] namely, octanoic acid ($C_8$), nonanoic acid ($C_9$), decanoic acid ($C_{10}$), and dodecanoic acid ($C_{12}$), since these sodium salt-based ESs can be seen as a result of replacing one of the acids by its sodium salt. The viscosity values for the sodium salt-based ES are higher than those composed exclusively of fatty acids. For example, comparing ESs composed of NaC10 and those composed of dodecanoic acid, with the same fatty acids, C8, C9, and C10 as HBD, and at the same composition $x_{acid}$ = 0.7, it can be seen that the viscosity values for the last ones vary between 7 and 11 mPa·s at 298.15 K, while for the former, they range between 209 and 226 mPa·s at the same temperature. It can be concluded that as expected, the introduction of the sodium salt increases the viscosity of these systems, probably due to the introduction of charges and thus electrostatic interactions.

An additional comparison can be performed for both density and viscosity between the sodium salt-based ESs studied here and those of the pure acids. The values of density and viscosity for the pure *n*-octanoic, *n*-nonanoic, and *n*-decanoic acids can be found in Table S7 of the Supporting Information. As expected, for the same temperature, the value of both these properties is higher for the ES, regardless of the composition, than for the corresponding pure acids. The density values of both sodium salt-based ESs, ranging from 0.961 to 0.979 g·cm$^{-3}$ for NaC6-based ESs and from 0.928 to 0.941 g·cm$^{-3}$, are slightly higher than those of pure acids, which are between 0.893 and 0.910 g·cm$^{-3}$. However, the viscosity values of the sodium salt-based ESs are much higher than those of the pure acids, indicating the establishment of strong or/and extensive intermolecular interactions between both components of the mixture. For example, the viscosity of NaC6/C9 is 280.5 mPa·s and of NaC12/C9 is 68.61 mPa·s at 293.15 K while that of the pure acid at the same temperature is only 8.08 mPa·s.

The activation energy, $E_a$, was calculated based on the viscosity dependence with temperature as follows

$$E_a = R\left(\frac{B_\eta}{\left(\frac{C_\eta^2}{T^2} - \frac{2C_\eta}{T} + 1\right)}\right) \tag{4}$$

where $\eta$ is the viscosity, $T$ is the temperature in K, $B_\eta$ and $C_\eta$ are the adjustable parameters, and $R$ is the universal gas constant. The analysis of the activation energy values allows the discussion of the energy barrier of a fluid to shear stress, since the higher the $E_a$, the more the difficult it is for the molecules/aggregates to move past each other, as a consequence of the size or entanglement of the molecules/aggregates and/or the presence of strong interactions within the fluid.

The values of the activation energies for the sodium salt-based ESs prepared in this work are presented in Table S8 in the Supporting Information. The same trend observed in viscosity values is also observed for the $E_a$ of dried ESs: $NaC_{12}/C_8$ acid < $NaC_{12}/C_9$ acid < $NaC_{12}/C_{10}$ acid. This trend shows that as the alkyl chain of acid increases, the $E_a$ of the ES increases in a linear manner.

**3.3. Solvatochromic Probes.** The polarity of the sodium salt-based ESs prepared in this work was measured using two polarity scales, the betaine dye scale, through the calculation of the solvatochromic parameters $E_T(33)$ and $E_TN$, which measure the overall interactions between the solvent and the dye,[37] and the Kamlet−Taft parameters, $\alpha$, $\beta$, and $\pi^*$, which


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measure the ability of the DES to donate and accept hydrogen bonds and polarizability, respectively. Since the experimental procedure followed was described in detail in the literature,[38] only the general procedure was explained here. In what concerns the betaine dye scale, the solvatochromic parameter $E_T(33)$ was determined from the lowest energy of charge-transfer absorption band of the dye in the sodium salt-based ESs under study using the following equation

$$E_T(33) = \frac{28591}{\lambda_{max}} \quad (5)$$

where $E_T(33)$ is in kcal·mol$^{-1}$ and $\lambda_{max}$ (nm) is the wavelength corresponding to the maximum of absorption of the probe in the solvent under study.[39] Absorbance spectra of Reichardt's 33 in the two families of sodium salt-based DESs were collected at room temperature, and the corresponding values of $E_T(33)$ were converted into $E_T(30)$ values, using the following equation[40,41]

$$E_T(30) = 0.9953(\pm 0.0287)E_T(33) - 8.1132(\pm 1.6546) \quad (6)$$

Then, the $E_TN$ normalized polarity parameter was calculated according to the following equation

$$E_TN = \frac{E_T(30) - 30.7}{32.4} \quad (7)$$

In what concerns the Kamlet–Taft scale, the 4-nitroaniline/*N*,*N*-diethyl-4-nitroaniline probe set was used. The $\beta$ values for the sodium salt-based DESs under study were calculated using the following equations

$$\beta = \frac{1.035\bar{\upsilon}_{N,N\text{-diethyl-4-nitroaniline}} + 2640 - \bar{\upsilon}_{4\text{-nitroaniline}}}{2800} \quad (8)$$

where $\bar{\upsilon}_{N,N\text{-diethyl-4-nitroaniline}}$ and $\bar{\upsilon}_{4\text{-nitroaniline}}$ are the maximum wave number of *N*,*N*-diethyl-4-nitroaniline and 4-nitroaniline, respectively, in kK.

The Kamlet–Taft $\alpha$ parameter provides a measure of a solvent's hydrogen-bond-donating acidity and was calculated using Reichardt's dye $E_T(30)$ parameter and the $\pi^*$ parameter as follows

$$\alpha = 0.0649 \times E_T(30) - 2.03 - 0.72\pi^* \quad (9)$$

Finally, the $\pi^*$ parameter, which provides a measure of the dipolarity and polarizability of the solvent, was calculated from spectroscopic data of the *N*,*N*-diethyl-4-nitroaniline probe in the sodium salt-based DESs through the following equation

$$\pi^* = 3.14 \times 10^{-4}(27.52 - \bar{\upsilon}_{N,N\text{-diethyl-4-nitroaniline}}) \quad (10)$$

where $\bar{\upsilon}$ is used in kK.

The calculated betaine dye parameters and the Kamlet–Taft solvatochromic parameters for the two families of sodium salt-based ESs prepared in this work are given in Table 4. The maximum absorption wavelengths for each solvent can be found in Table S9 in the Supporting Information and the UV–vis spectra are included in Figures S3–S20 in the Supporting Information.

No significant differences were observed for the solvatochromic parameter $E_TN$ between the two families of sodium salt-based DESs, which is not surprising since these two families of DESs are not chemically significantly different from each other in terms of polarity, even with different proportions of HBDs. However, the $E_TN$ values for the NaC12-based family of DESs are slightly lower than the homologous DES in the NaC6-based family. Also, a slight decrease in the $E_TN$ with the increase in the alkyl chain of the fatty acid HBD was observed.

**Table 4. Betaine Dye and Kamlet–Taft Solvatochromic Parameters Using a Dye Set of 4-Nitroaniline, *N*,*N*-Diethyl-4-Nitroaniline, and Reichardt's Dye for the Two Families of Sodium Salt-Based DESs Under Study**

| solvent | $\alpha$ | B | $\pi^*$ | $E_T(33)$ | $E_T(30)$ | $E_TN$ |
|---|---|---|---|---|---|---|
| NaC6/C8 | 1.48 | 0.50 | 0.38 | 66.65 | 58.22 | 0.85 |
| NaC6/C9 | 1.47 | 0.47 | 0.38 | 66.55 | 58.13 | 0.85 |
| NaC6/C10 | 1.44 | 0.43 | 0.38 | 66.12 | 57.70 | 0.83 |
| NaC12/C8 | 1.44 | 0.41 | 0.38 | 65.99 | 57.56 | 0.83 |
| NaC12/C9 | 1.45 | 0.38 | 0.37 | 66.18 | 57.76 | 0.84 |
| NaC12/C10 | 1.42 | 0.38 | 0.35 | 65.48 | 57.06 | 0.81 |

In Figures 6 and 7, the obtained results for the $E_TN$ solvatochromic parameter for the sodium salt-based DESs are compared with those of other DESs from the literature.[38]

It can be observed that $E_T$N results for NaC6- and NaC12-based ESs are similar to those obtained for choline chloride-based DESs and higher than those reported for tetraalkylammonium chloride-based DESs. A direct comparison of the effect of the HBA can be made using DESs with octanoic acid as HBD N4444Cl/C8 (0.69)[42] < NaC6/C8 (0.85). This clearly shows the important role of HBAs in defining the overall ES polarity. The lower polarity of the tetraalkylammonium chloride-based family of DESs can probably be explained by the cation charge screening due to the presence of the alkyl chains, whereas this phenomenon is completely absent in the NaC6- and NaC12-based ESs. The explicit charge of the sodium atom is probably responsible for high $E_TN$ values for these salts, conferring a similar polarity level of hydrophilic and polar choline chloride-based DESs.

In Figure 7, the values of $\alpha$, $\beta$, and $\pi^*$ Kamlet–Taft solvatochromic parameters for the two families of sodium salt-based ESs prepared in this work are compared with those of other DESs taken from the literature.[42] It can be observed that the values of the solvatochromic parameters, especially of $\alpha$ and $\pi^*$, do not differ significantly within each DES family. For example, for NaC6/C10 ES, $\alpha$ and $\pi^*$ are 1.44 and 0.38, while for NaC12/C10 ES, these parameters take the value of 1.42 and 0.35. However, these results are similar to those reported for choline chloride-based DESs but smaller than those reported for tetraalkylammonium chloride-based DESs. Again, a direct comparison of the effect of the HBA in the $\alpha$ parameter using DESs with octanoic acid as an HBD, N4444Cl/C8 (0.98)[42] < NaC6/C8 (1.35), clearly shows the important role of HBAs in the hydrogen-bonding/accepting properties of DESs. Within each family, now regarding the effect of HBD on the $\alpha$ parameter, a slight decrease can be observed for both NaC6- and NaC12-based ES families, which is in agreement with what was observed by other authors, for example, for the tetraalkylammonium chloride-based DESs.

The $\beta$ parameter shows the studied ES's capability to accept hydrogen bonds. It can be observed that the values of this parameter are slightly smaller for the NaC6-based ES family than for NaC12-based ESs. It can also be observed that, within each family, there is a decrease in $\beta$ parameter with the increase in the alkyl chain length of the HBD. This decrease in $\beta$ parameter with the alkyl chains of both the HBA and HBD is


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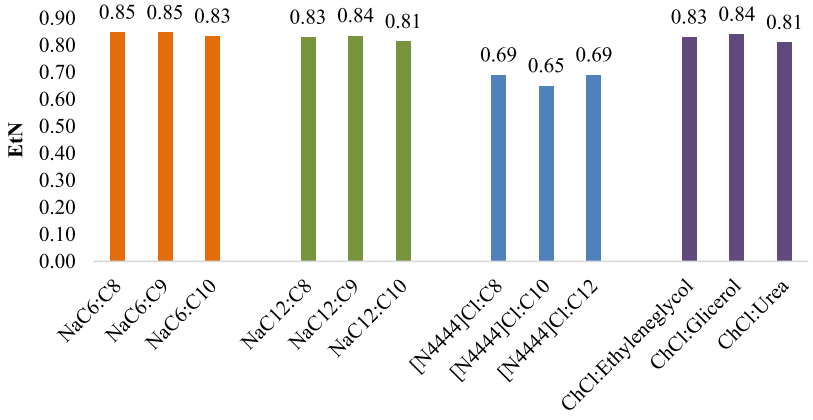



**Figure 6.** Comparison of the $E_TN$ betaine dye solvatochromic parameter for the sodium salt-based ESs studied in this work with literature results for ammonium salt-based DESs.[10] The compositions $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 were chosen, for comparison purposes, for NaC6- and NaC12-containing ES families, respectively. The HBA/HBD proportions of the literature DES are (1:2).

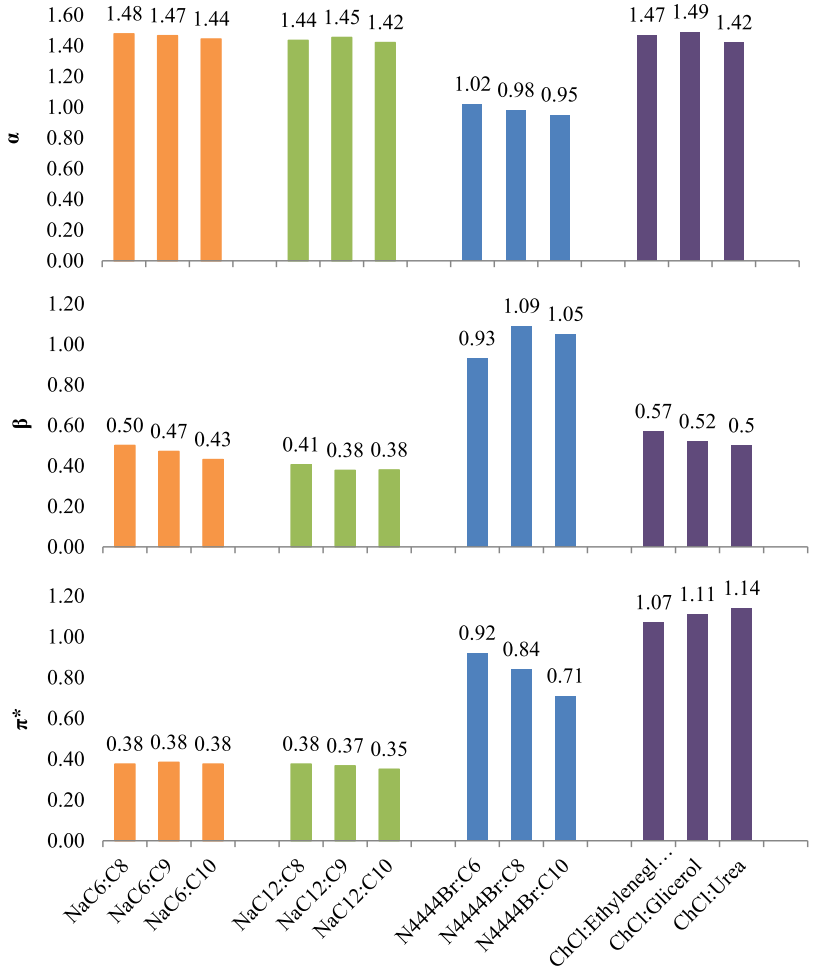



**Figure 7.** Comparison of the Kamlet−Taft parameters, $\alpha$, $\beta$, and $\pi^*$, for the sodium salt-based ESs studied in this work with literature results for ammonium salt-based DESs.[38,42] The compositions $x_{acid}$ = 0.7 and $x_{acid}$ = 0.8 were chosen, for comparison purposes, for NaC6- and NaC12-containing ES families, respectively. The HBA/HBD proportions of the literature DESs are (1:2).

in agreement with the basicity order of these compounds and thus can be explained by the stronger interactions between the HBD and HBA as the alkyl chain becomes smaller and thus their inability to interact with the probe. The effect of the HBA in the $\beta$ parameter is more pronounced than that of the HBD and can clearly be evaluated by comparing the values of $\beta$ for the three families of compounds: sodium salt-based ESs ≈ choline chloride-based DESs ≫ N4444Br-based DESs. The similar values for the first two families indicate their higher localized charged character of the HBA in comparison with the N4444Br family where the alkyl chains in the HBA shield the charge present in the ammonium cation.

As for the $\pi^*$ parameter, very similar values were obtained for the two sodium salt-based families of ESs and much smaller



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than those obtained for other DESs in the literature, as the values are close to a third of those of the other DESs. This means that the HBA has a crucial role in the polarizability sodium salt-based ES molecules as they are much less polarizable than the $N_{4444}Br$ salt, due to the presence of a single alkyl chain present in the salt, making it more rigid and thus less polarizable. It can also be seen that for the NaC12-based family that there is a slight tendency for the $\pi^*$ to decrease with the increase in the alkyl side chain, which was also observed for the N4444Br family of DESs. A noncharged hydrophobic DES has lower $\pi^*$ and higher $\beta$ values than a more hydrophilic DES. This is not true here due to the presence of the sodium atom that confers a more polar character to the ES, enabling the incorporation of water and the formation of gels.[27]

## 4. CONCLUSIONS

In this work, novel DESs were developed, containing sodium salts derived from long alkyl chain carboxylic acids and long alkyl chain carboxylic acids, of which the former act as HBAs and the latter as HBDs. These new solvents are composed of sodium hexanoate and sodium dodecanoate, as well as octanoic, nonanoic, and dodecanoic acid.

The solid−liquid phase diagrams of the ES were determined using turbidimetry and DSC. The location of the eutectic point range of compositions allows us to conclude that the HBA has a stronger influence in the composition of the eutectic point. Measurements of thermophysical properties, such as density and viscosity, between 293.15 and 353.15 K and at atmospheric pressure, were carried out. These ESs present densities, at 298.15 K, from 0.957 to 0.975 g·cm$^3$, and NaC12-based ESs present densities, ranging from 0.924 to 0.938 g·cm$^3$, respectively. Despite their charged nature and their long chains in both the HBA and HBD, they present similar viscosities, comparing hydrophobic DESs based on ammonium salts with long alkyl chains.

Finally, polarity of the sodium salt-based ES was measured using solvatochromic probes and the polarity parameters ($E_T$ and $E_TN$) and Kamlet−Taft parameters ($\alpha$, $\beta$, and $\pi^*$) were obtained. The values obtained for these parameters are in between those of hydrophilic and hydrophobic DESs, with low polarizability and hydrogen-bond-donating character and high hydrogen-bonding/accepting capacity.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jced.1c00188.

Experimental data of solid−liquid phase equilibria, DSC thermograms, density and viscosity of dried sodium salt-based DESs and their derived properties, chemical structure of the solvatochromic probes used, and UV−vis spectra (PDF)

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### Notes

The authors declare no competing financial interest.

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